Linux day-2

```
[sharafkamaal@192 ~ % chmod 777 kamaalz123
[sharafkamaal@192 ~ % ls
 amzon-affiliate-n8n.json
 Applications
 cascade-projects
 clawd
 Desktop
 devops-techiehorizon
 docker
 Documents
 Downloads
 dyad-apps
 dyad-darwin-arm64-0.35.0.zip
 dyad.app
 helloai
 ideogram-v3.0_Create_a_4K_ultra-premium_SaaS_comparison_poster_
 ideogram-v3.0_Create_a_luxury_4K_ultra-high-resolution_infograp
 kamaalz123
```

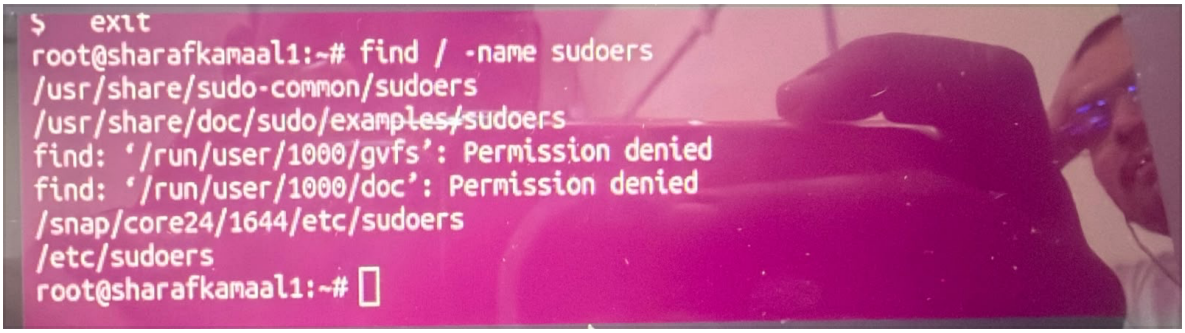
chmod 777 <target>: Grants full read, write, and execute permissions (rwxrwxrwx) to the owner, group, and all other users on the specified file or directory.

```
[sharafkamaal@192 ~ % cat /etc/passwd
##
# User Database
#
# Note that this file is consulted directly only when the system is running
# in single-user mode.  At other times this information is provided by
# Open Directory.
#
# See the opendirectoryd(8) man page for additional information about
# Open Directory.
##
nobody:*:-2:-2:Unprivileged User:/var/empty:/usr/bin/false
root:*:0:0:System Administrator:/var/root:/bin/sh
daemon:*:1:1:System Services:/var/root:/usr/bin/false
_uucp:*:4:4:Unix to Unix Copy Protocol:/var/spool/uucp:/usr/sbin/uucico
_taskgated:*:13:13:Task Gate Daemon:/var/empty:/usr/bin/false
_networkd:*:24:24:Network Services:/var/networkd:/usr/bin/false
_installassistant:*:25:25:Install Assistant:/var/empty:/usr/bin/false
_lp:*:26:26:Printing Services:/var/spool/cups:/usr/bin/false
_postfix:*:27:27:Postfix Mail Server:/var/spool/postfix:/usr/bin/false
_scsd:*:31:31:Service Configuration Service:/var/empty:/usr/bin/false
_ces:*:32:32:Certificate Enrollment Service:/var/empty:/usr/bin/false
_appstore:*:33:33:Mac App Store Service:/var/db/appstore:/usr/bin/false
_mcxalr:*:54:54:MCX AppLaunch:/var/empty:/usr/bin/false
_appleevents:*:55:55:AppleEvents Daemon:/var/empty:/usr/bin/false
_geod:*:56:56:Geo Services Daemon:/var/db/geod:/usr/bin/false
_devdocs:*:59:59:Developer Documentation:/var/empty:/usr/bin/false
_sandbox:*:60:60:Seatbelt:/var/empty:/usr/bin/false
_mdnsresponder:*:65:65:mDNSResponder:/var/empty:/usr/bin/false
_ard:*:67:67:Apple Remote Desktop:/var/empty:/usr/bin/false
_www:*:70:70:World Wide Web Server:/Library/WebServer:/usr/bin/false
_eppc:*:71:71:Apple Events User:/var/empty:/usr/bin/false
_cvs:*:72:72:CVS Server:/var/empty:/usr/bin/false
_svn:*:73:73:SVN Server:/var/empty:/usr/bin/false
_mysql:*:74:74:MySQL Server:/var/empty:/usr/bin/false
_sshd:*:75:75:sshd Privilege separation:/var/empty:/usr/bin/false
```

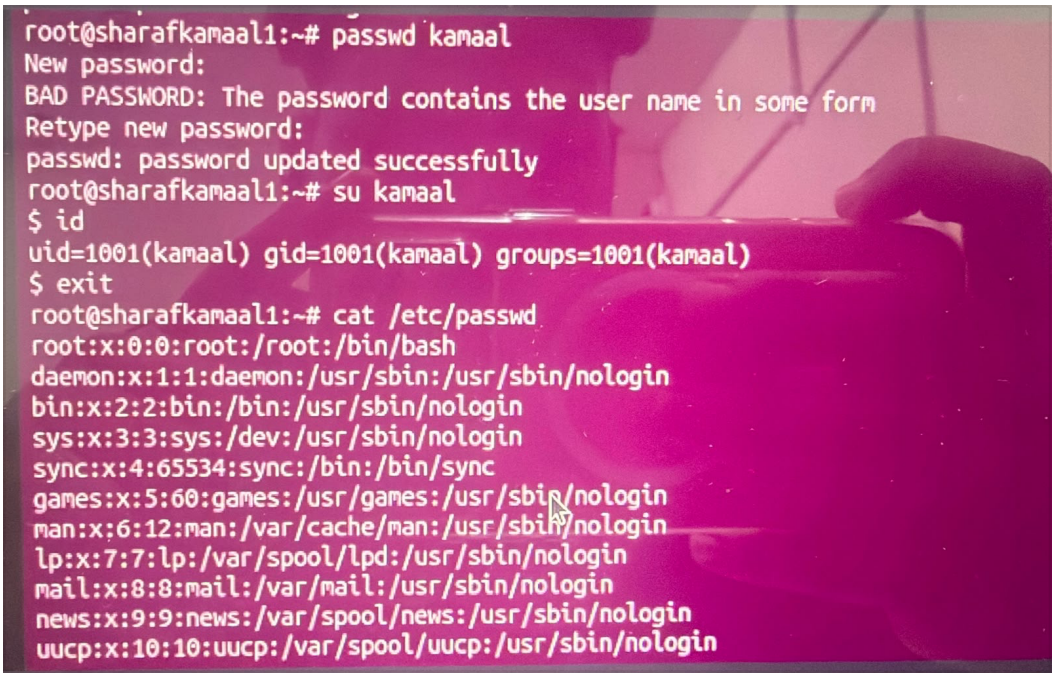
cat /etc/passwd: Displays the contents of the /etc/passwd file, listing all registered local user accounts on the system along with their User IDs (UID), Group IDs (GID), home directories, and default shells.

```
[sharafkamaal@192 ~ % chown root:root kamaalz123
 chown: root: illegal group name
[sharafkamaal@192 ~ % sudo chown root:root kamaalz123
[Password:
 chown: root: illegal group name
[sharafkamaal@192 ~ % sudo chown root kamaalz123
[sharafkamaal@192 ~ % ls -lh
 total 956896
 -rw-r--r--@   1 sharafkamaal  staff    15K  1 Mar  2026 amzon-affilia
 drwx------@   5 sharafkamaal  staff   160B 25 Jan  2026 Applications
 drwxr-xr-x   14 sharafkamaal  staff   448B 15 Jun 18:54 cascade-proje
 drwxr-xr-x@  15 sharafkamaal  staff   480B  5 Feb  2026 clawd
 drwx------@  25 sharafkamaal  staff   800B  9 Oct 11:56 Desktop
 drwxr-xr-x    7 sharafkamaal  staff   224B  9 Oct 11:40 devops-techie
 drwxr-xr-x@   4 sharafkamaal  staff   128B 11 Jan  2026 docker
 drwx------@   6 sharafkamaal  staff   192B 12 Mar  2026 Documents
 drwx------@ 168 sharafkamaal  staff   5.3K  9 Oct 11:41 Downloads
 drwxr-xr-x@   2 sharafkamaal  staff    64B  5 Feb  2026 dyad-apps
 -rw-r--r--@   1 sharafkamaal  staff   179M  5 Feb  2026 dyad-darwin-a
 drwxr-xr-x@   3 sharafkamaal  staff    96B  3 Feb  2026 dyad.app
 drwxr-xr-x    3 sharafkamaal  staff    96B 10 Jan  2026 helloai
 -rw-r--r--@   1 sharafkamaal  staff   992K 24 Feb  2026 ideogram-v3.0
 ck-0.jpg
 -rw-r--r--@   1 sharafkamaal  staff   1.0M 24 Feb  2026 ideogram-v3.0
 -0.jpg
 dr--rwx--x    4 root          staff   128B  8 Oct 16:10 kamaalz123
```

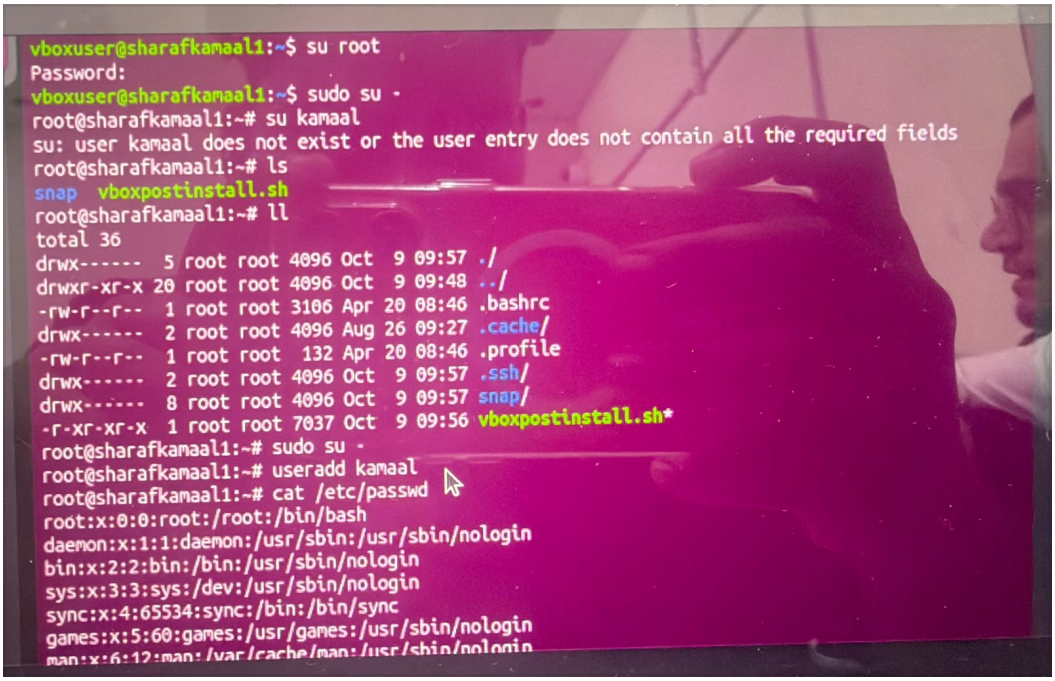
- **chown root:root <file>**: Changes both the owner user and the owner group of the specified file to root.
- **sudo chown root:root <file>**: Executes the user-and-group ownership change to root with elevated administrative (superuser) privileges.
- **ls -lh**: Lists directory contents in detailed long format (-l) with file sizes shown in human-readable units (-h, like KB, MB, or GB).



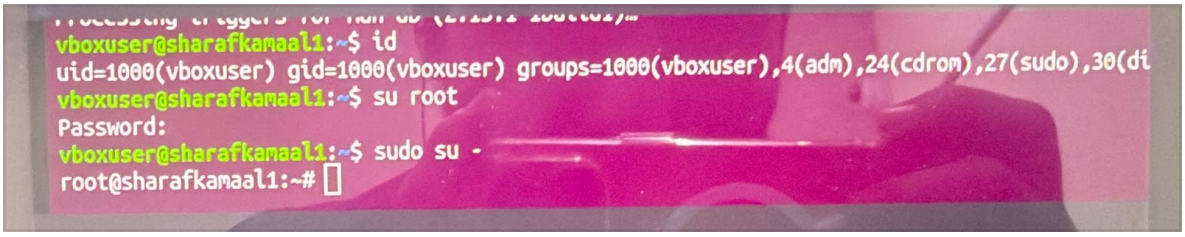
- find / -name sudoers: Searches the entire filesystem starting from the root directory (/) for any file or directory named exactly sudoers.



- **passwd**: Changes the current user's password, or allows a superuser (sudo passwd <user>) to set or modify another account's password.



su root: Switches to the root user account by prompting for the root password, but keeps your current user's shell environment and path.
sudo su -: Switches to the root user using your own sudo privileges (requiring your password, not root's) with a hyphen (-) to load a fresh, clean login shell environment identical to logging in directly as root.



id: Displays the user identity details